
ReasonMotion: Language-Guided Editing and Reward-Guided Prediction for Sports Coaching

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Language-guided human motion modeling has advanced rapidly, but sports coach-
2 ing poses a more demanding problem: a useful system must support fine-grained
3 technical edits and predict how those edits affect downstream motion under biome-
4 mechanical constraints. Existing approaches are largely designed for general human
5 motion and trained with supervised imitation, which limits both localized con-
6 trollability and adaptation toward higher-quality athletic outcomes. We present
7 ReasonMotion, a unified framework for sports coaching that combines language-
8 guided editing with reward-guided prediction. Its Spatial-Temporal-Aware Mixture-
9 of-Experts (ST-MoE) architecture decomposes motion modeling into specialized
10 temporal and spatial experts for precise instruction grounding, while visual Group
11 Relative Policy Optimization (vis-GRPO) optimizes diffusion-based trajectories
12 toward sport-specific objectives beyond imitation learning. We evaluate ReasonMo-
13 tion on MotionFix for motion editing, Human3.6M for general motion prediction,
14 and FineFS for figure-skating coaching scenarios. ReasonMotion performs strongly
15 on instruction-grounded motion editing, competitive results on Human3.6M, and
16 a 21.4% F-MPJPE improvement over the diffusion baseline. Moreover, expert
17 evaluation by coaches and athletes further confirms our outputs are preferred in
18 technical quality, biomechanical plausibility, and coaching utility, establishing
19 ReasonMotion as an effective framework for sports coaching-oriented motion
20 generation that shifts the paradigm from simple synthesis to consequence-aware
21 optimization.

22 1 Introduction

23 Human motion generation plays a pivotal role in a wide range of applications, from virtual reality to
24 sports training and robotics [45, 15, 5, 43]. With advances in multimodal learning, recent work has
25 explored using natural language to guide human motion. This task, typically referred to as text-to-
26 motion (T2M), is opening up new possibilities in cross-modal generation, controllable animation,
27 and motion synthesis [7, 42, 18]. However, applying these advances to competitive sports requires a
28 fundamental shift in perspective: a coach’s instruction is a causal intervention that propagates through
29 the kinetic chain, and validating any edit requires predicting its downstream consequences—mirroring
30 real coach–athlete feedback.

31 Existing T2M frameworks struggle with this cause-and-effect relationship for two reasons. First,
32 most models focus on coarse stylistic transformations (e.g., walking to dancing) and lack the capacity
33 for precise, localized adjustments (e.g., “bend torso forward during jump”) required in technical
34 sports. Secondly, these models are designed for general human motion and trained with supervised
35 learning. Consequently, they converge on the average motion patterns of the training distribution
36 rather than adapting motion toward higher-scoring athletic patterns. In competitive training, the goal
37 is not to perform a likely motion, but to identify the specific kinetic adaptation that maximizes the

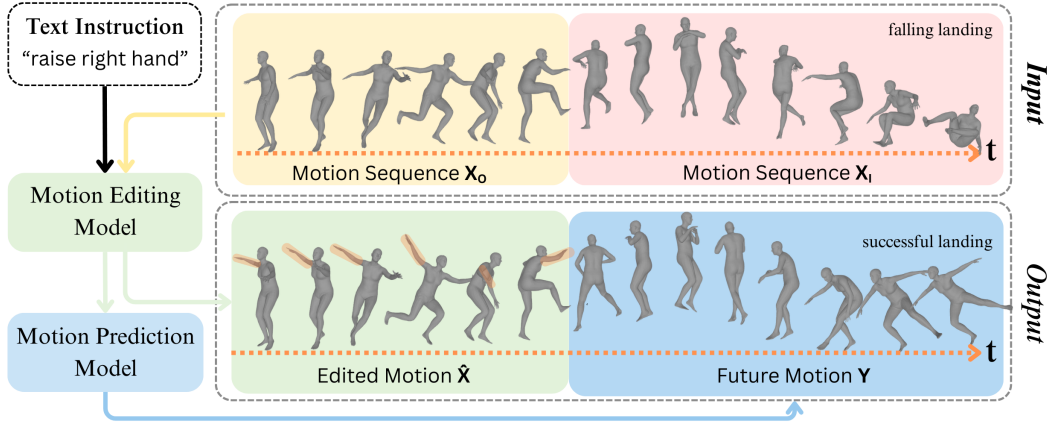


Figure 1: **An illustration of the ReasonMotion framework.** ReasonMotion acts as a biomechanical “what-if” simulator. By grounding a coaching constraint (e.g., “raise right hand”) the model reasons through the kinetic chain adaptations necessary to maximize the final athletic score.

38 final outcome. To model this complex dynamic, we propose ReasonMotion, a unified framework built
 39 on a novel biomechanical reasoning process. Instead of imitating average behaviors, ReasonMotion
 40 models how localized interventions propagate through the kinetic chain. It answers the fundamental
 41 coaching question: *If an athlete enters a movement with a specific technical constraint, what plausible*
 42 *continuation best satisfies the technical constraint while improving the predicted performance score?*
 43 To achieve this, we first propose the Spatial-Temporal-Aware Mixture-of-Experts (ST-MoE), an
 44 architecture that provides the foundation for understanding by deconstructing complex motion into
 45 manageable spatio-temporal modules to accurately isolate and apply the coach’s localized semantic
 46 edit.

47 We further introduce visual Group Relative Policy Optimization (*vis*-GRPO) to refine prediction with
 48 sport-specific rewards, shifting learning from imitation to outcome optimization. Unlike supervised
 49 imitation, *vis*-GRPO performs group-relative comparisons to drive stable policy updates. Crucially,
 50 *vis*-GRPO updates the ST-MoE routing gates, enabling the model to select expert pathways that yield
 51 higher-scoring biomechanical adaptations under the same constraint.

52 Our main contributions are summarized as follows:

- 53 • We propose ReasonMotion, a unified framework for language-guided motion editing and
 54 reward-guided motion prediction for sports coaching.
- 55 • We introduce ST-MoE, a spatio-temporal mixture-of-experts architecture that improves fine-
 56 grained instruction grounding by decomposing motion modeling into specialized temporal
 57 and spatial experts.
- 58 • We introduce *vis*-GRPO, a reward-guided optimization method for diffusion-based motion
 59 prediction that encourages trajectories aligned with sport-specific performance objectives
 60 beyond supervised imitation.
- 61 • We demonstrate strong empirical performance across both editing and prediction bench-
 62 marks, including improved instruction-grounded editing on MotionFix, competitive ADE
 63 performance on Human3.6M, and a 21.4% F-MPJPE improvement over the diffusion base-
 64 line on FineFS.
- 65 • We provide domain-expert human evaluation with professional figure skating coaches and
 66 competitive athletes, showing that ReasonMotion’s outputs are preferred in technical quality,
 67 biomechanical plausibility, and coaching utility.

68 2 Related Work

69 **General motion generation.** Recent advances in human motion generation and prediction have
70 introduced a variety of text- or sequence-guided models that demonstrate strong general-purpose
71 performance [8, 40, 3]. For instance, Back to MLP [10] presents a compact MLP-based framework
72 for 3D motion prediction; Light-T2M [39] combines Mamba and textual injection for efficient
73 text-conditioned motion generation, while MotionDiffuse [41] employs motion VAE and classifier-
74 free guidance to synthesize realistic and diverse motions. Despite their effectiveness on general
75 motion, these models lack fine-grained controllability or domain adaptation essential for precise,
76 sport-oriented motion synthesis. Other recent works such as MotionFix [1] and MotionLab [11]
77 further enhanced controllability within motion generation frameworks, bridging the gap between
78 free-form synthesis and targeted editing. Despite these advances, these methods remain limited in
79 capturing semantic and biomechanical subtleties critical to athletic motions.

80 **Sport-specific motion generation.** Several works further explore sport-specific motion generation,
81 addressing the unique biomechanical and reasoning challenges of athletic movements. Video-based
82 methods such as ViMo [21] leverage visual cues from sports footage but are constrained to video-to-
83 motion generation. Diffusion-based control models like SMGDiff [34] achieve real-time performance
84 but remain domain-restricted (soccer-only), while trajectory-oriented approaches such as Sports-Traj
85 [33] emphasize spatial coordination over joint-level precision. While these methods improve realism
86 within sport domains, they still lack the generalizable reasoning and joint-level control required for
87 complex athletic motion generation.

88 **Reasoning-augmented motion generation.** Reasoning-augmented motion generation leverages
89 LLMs or structured prompting to incorporate semantic understanding into motion synthesis. Mo-
90 tionGPT [22] prompts low-level textual descriptions for natural motion generation but relies heavily
91 on prompt alignment, limiting adaptability in precision-critical sports. HUMANISE [3] produces
92 context-aware motions; however, it is limited to simple indoor scenarios. AvatarGPT [44] and
93 Motion-Agent [32] introduce LLM-driven planning and motion-text interaction, yet they abstract
94 away fine-grained kinematics and remain insufficiently accurate for athletic motions. Although these
95 approaches improve contextual awareness, their “reasoning” is largely confined to the semantic
96 text space. They are not designed for technical sports, where reasoning must involve physical,
97 biomechanical adaptation to maximize a performance reward.

98 ReasonMotion uniquely bridges the gap between these domains. Unlike prior general, domain-
99 specific, or LLM-based models, our framework specifically simulates an athlete-coach interaction.

100 3 Methods

101 The ReasonMotion architecture consists of two core components: the ST-MoE architecture to provide
102 the decomposable structure, and the vis-GRPO algorithm to guide the model toward high-performance
103 outcomes. The core of our reasoning process is their synergy: vis-GRPO provides the performance-
104 driven learning signal that trains the ST-MoE’s expert routing gates. This process teaches the model
105 to reason by selecting the optimal combination of experts to maximize a reward signal, rather than
106 just mimicking ground-truth data. The architecture of ReasonMotion is illustrated in Figure 2.

107 3.1 Conditional-Text Diffusion

108 We adopt Temporal Cascaded Diffusion (TCD) [23] as our backbone due to its spatio-temporal in-
109 painting design, which naturally supports motion editing while preserving sequence continuity. Since
110 the original TCD is coordinate-driven and does not incorporate language guidance, we extend each
111 ResidualBlock with text cross-attention. Specifically, text embeddings from a lightweight sentence-
112 transformer [30] serve as the Key and Value, while the diffusion-time-aware motion representation
113 $(X + t)$ serves as the Query. The cross-attention output is added through a residual connection before
114 being passed to the time and feature self-attention layers, yielding a text-conditional diffusion model
115 for instruction-guided motion generation.

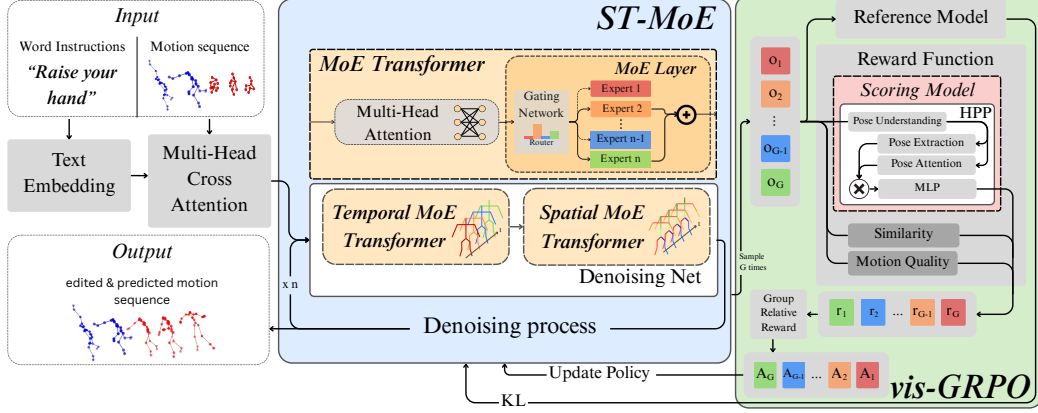


Figure 2: The ReasonMotion architecture consists of a ST-MoE diffusion model for motion generation and a vis-GRPO module for reward-guided refinement.

3.2 Reasoning Process

To implement the reasoning process outlined above, we first detail the ST-MoE architecture, followed by the vis-GRPO algorithm.

3.2.1 ST-MoE

As the architectural foundation of our reasoning process, we propose ST-MoE. We are inspired by the Mixture-of-Experts (MoE) paradigm [26], which has been used in models like DeepSeek-V3 [14] to enhance reasoning and reduce computational cost. However, the MoE structure is not curated for the unique demands of human motion. In motion generation and understanding, the model must have the ability to decompose a complex, continuous action into manageable, specialized parts.

ST-MoE overcomes this challenge by processing the motion representation along two distinct aspects. The **Temporal MoE** layer focuses on modeling joint-wise trajectories over time, allowing its experts to capture dynamic patterns such as acceleration and periodicity. In contrast, the **Spatial MoE** layer captures dependencies across joints at each frame, enabling expert specialization over body parts or coordinated structures (e.g., limbs vs. torso). ST-MoE is implemented within each residual block as a sequential composition of two specialized MoE Transformer layers. Given the text-conditioned motion tensor $X \in \mathbb{R}^{B \times C \times K \times L}$ (Batch, Channels, Joints, Frames), the data flows as:

$$Y = \text{MoE-Transformer}_{\text{Time}}(X), \quad (1)$$

$$Z = \text{MoE-Transformer}_{\text{Space}}(Y). \quad (2)$$

Each MoE-Transformer is a standard Transformer Encoder Layer where the Feed-Forward Network (FFN) is replaced by a sparse MoE FFN with N expert networks and a learned router $W_r \in \mathbb{R}^{N \times d}$. Given an input token x , the router computes logits $s = W_r x$ and selects the Top- K experts:

$$\mathcal{I} = \text{TopK}(\text{softmax}(s)), \quad (3)$$

$$g_i = \frac{\exp(s_i)}{\sum_{j \in \mathcal{I}} \exp(s_j)}, \quad i \in \mathcal{I}, \quad (4)$$

$$\text{MoE-FFN}(x) = \sum_{i \in \mathcal{I}} g_i \cdot E_i(x), \quad (5)$$

where $E_i(\cdot)$ denotes the i -th expert network and the gate weights g_i are renormalized over only the selected experts.

3.2.2 vis-GRPO

To optimize the ST-MoE policy π_θ , we introduce vis-GRPO, a variant of the GRPO algorithm [25] adapted for visual motion generation and trajectory prediction. Current models rely on supervised

learning, which restricts them to imitating ground-truth data. This approach lacks the ability to explore novel, high-scoring motion strategies crucial in sports. Our vis-GRPO reasoning process addresses this by using reinforcement learning to continuously improve motion quality based on performance rewards. Inspired by the success of GRPO in language models [9], vis-GRPO leverages relative comparisons within a generated group to provide a clear and stable learning signal. The policy optimization process is defined by:

$$J_{\text{vis-GRPO}}(\theta) = \mathbb{E}_{\substack{q \sim P(Q) \\ \{o_i\} \sim \pi_{\theta_{\text{old}}}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)} A_i, \right. \right. \\ \left. \left. \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) \right] - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (6)$$

Here, q represents the input query (e.g., observed motion and text) and $\{o_i\}_{i=1}^G$ is a group of G output motions sampled from the old policy $\pi_{\theta_{\text{old}}}$. The term ϵ is the clipping parameter. The advantage A_i is computed using group-wise normalization: $A_i = (r_i - \text{mean}(\mathbf{r})) / (\text{std}(\mathbf{r}) + \delta)$, where r_i is the reward for the i -th sample. Finally, $D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$ is a KL divergence penalty with coefficient β against a reference policy π_{ref} .

To compute the policy probabilities $\pi_{\theta}(o_i|q)$ in diffusion models, we interpret the policy as the stochastic reverse denoising process that induces the final output motion. Specifically, each generated motion o_i is obtained from a denoising trajectory

$$\tau_i = (x_T^i, x_{T-1}^i, \dots, x_0^i), \quad o_i = x_0^i, \quad (7)$$

where the trajectory probability factorizes over reverse diffusion transitions:

$$\pi_{\theta}(\tau_i | q) = p(x_T^i | q) \prod_{t=T}^1 p_{\theta}(x_{t-1}^i | x_t^i, q). \quad (8)$$

Therefore, $\pi_{\theta}(o_i|q)$ in Eq. 6 is implemented through the likelihood of the denoising trajectory that produces o_i .

Each reverse transition is modeled as a Gaussian distribution with fixed variance:

$$p_{\theta}(x_{t-1}^i | x_t^i, q) = \mathcal{N}(\mu_{\theta}(x_t^i, t, q), \sigma_t^2 \mathbf{I}), \quad (9)$$

where $\mu_{\theta}(x_t^i, t, q)$ is the reverse-process mean induced by the diffusion model prediction, and $\sigma_t^2 \mathbf{I}$ is a fixed diagonal covariance. Following [28] and [20], this fixed-variance Gaussian formulation is sufficient to represent human motion and allows the L2 distance to serve as a tractable proxy for the evidence lower bound (ELBO), ensuring the vis-GRPO objective remains grounded in maximizing data log-likelihood.

The primary novelty of **vis-GRPO** lies in computing policy ratios over continuous diffusion transitions. Unlike in language modeling, where the sequence probability factorizes into a product of discrete token probabilities, our policy ratio is computed over Gaussian reverse-process densities, providing a stable and computationally efficient signal to identify which samples are “better” than the group average. In practice, the likelihood ratio in Eq. 6 is computed from the reverse-process transition densities. For each denoising step, we compute

$$\rho_{i,t}(\theta) = \exp(\log p_{\theta}(x_{t-1}^i | x_t^i, q) - \log p_{\theta_{\text{old}}}(x_{t-1}^i | x_t^i, q)). \quad (10)$$

By substituting the Gaussian transition policy in Eq. 9, the constant terms cancel out, and the step-wise log-ratio becomes a comparison of squared Euclidean distances:

$$\log \rho_{i,t}(\theta) = \frac{1}{2\sigma_t^2} (\|x_{t-1}^i - \mu_{\theta_{\text{old}}}(x_t^i, t, q)\|^2 - \|x_{t-1}^i - \mu_{\theta}(x_t^i, t, q)\|^2). \quad (11)$$

This formulation optimizes the policy π_{θ} to make its denoising transitions more likely for high-reward samples than the old policy. In implementation, rewards are assigned to the final generated motions $o_i = x_0^i$, while the GRPO likelihood ratios are computed from the per-step reverse diffusion log-probabilities along the sampled denoising trajectories.

175 **Rollout Routing Replay (R3) for GRPO** Training ST-MoE with GRPO introduces a routing
 176 inconsistency problem: the rollout and gradient passes may select different experts for the same input,
 177 causing the policy gradient to correspond to a path never traversed during sampling. We resolve this
 178 with R3 [16], which records the expert assignments $\{\mathcal{T}_t^{\text{infer}}\}$ during rollout and replays them during
 179 the training pass, ensuring gradient computation on the exact computational path that produced the
 180 trajectory. We provide the full formulation in Appendix A.

181 3.2.3 Reward Model

182 We design three domain-specific rewards combined as a weighted sum:

$$r = \lambda_{\text{score}} \cdot r_{\text{score}} + \lambda_{\text{similarity}} \cdot r_{\text{similarity}} + \lambda_{\text{quality}} \cdot r_{\text{quality}} \quad (12)$$

183 r_{score} assigns a scalar performance score via a two-stage network: a Human Pose Perception (HPP)
 184 module [36] encodes spatio-temporal skeletal features, followed by an MLP regression head trained
 185 with MSE against ground-truth labels. $r_{\text{similarity}}$ penalizes L2 deviation from ground-truth joint
 186 coordinates, and r_{quality} encourages physical plausibility via a smoothness term (penalizing velocity
 187 and acceleration fluctuations) and a bone consistency term (penalizing temporal variation in inter-joint
 188 distances). Full formulations are provided in Appendix B.

189 3.3 Motion Prediction Model

190 Given an observed motion prefix $x_{\text{past}} \in \mathbb{R}^{K \times L \times 3}$ and a semantic token S (e.g., “triple lutz”), our
 191 goal is to generate a physically coherent future trajectory $x_{\text{future}} \in \mathbb{R}^{K \times L' \times 3}$. This task directly
 192 instantiates our reasoning formulation in Sections 3.2.1–3.2.3: the predictor M_P is a text-conditional
 193 diffusion policy whose ST-MoE routing gates determine how spatio-temporal experts contribute to
 194 kinetic adaptation, while *vis*-GRPO refines these gates toward higher-reward outcomes. Conditioning
 195 on S resolves the inherent one-to-many ambiguity by constraining the feasible continuation to the
 196 intended motion type and difficulty.

197 **Stage 1: SSL (Initialization).** We train M_P with the standard diffusion L2 objective to reconstruct
 198 ground-truth x_{future} given (x_{past}, S) , establishing a strong prior over physically plausible continuations
 199 and a stable initialization for ST-MoE routing.

200 **Stage 2: *vis*-GRPO (Reward-guided routing).** Starting from the SSL model, we apply *vis*-GRPO
 201 where M_P serves as π_θ . For each query $q = (x_{\text{past}}, S)$, we sample G candidate futures and score
 202 them via the Reward Model (r_{quality} , r_{score} , and optionally $r_{\text{similarity}}$). *vis*-GRPO then updates π_θ to
 203 favor higher-reward candidates, training the ST-MoE gates to select expert pathways yielding better
 204 biomechanical adaptations under the same semantic constraint.

205 3.4 Motion Editing Model

206 Given an original motion $x_{\text{orig}} \in \mathbb{R}^{K \times L \times 3}$ and a text instruction S (e.g., “raise right hand”), our
 207 goal is to generate an edited motion $\hat{x}$ that preserves unedited regions while applying the specified
 208 modification. We use a simplified TCD variant with two input-level changes: the input is reduced to
 209 a single channel $(B, 1, K, L)$, and x_{orig} is appended as a side-conditioning signal at each denoising
 210 step [23], preventing trivial copying and enforcing genuine instruction-driven edits. As the editing
 211 task has a clear supervised signal, we train with the standard diffusion L2 objective without *vis*-GRPO.

212 4 Experiments

213 4.1 Datasets

214 We evaluate on three datasets, a sport-specific dataset and two general motion datasets. **FineFS** [35]
 215 is a figure skating benchmark of 1,167 video clips annotated with hierarchical action labels and Grade
 216 of Execution (GOE) scores (−5 to +5). We extract 4,584 jump segments as 24-joint skeletons via Hy-
 217 brIK [13], retaining half after quality filtering, and split into train/val/test at 8:1:1. **Human3.6M** [12]
 218 contains 3.6 million body poses across 15 action categories from seven actors, for which we adopt
 219 a 17-joint representation and follow the standard evaluation protocol [19]. **MotionFix** [1] provides
 220 6.7k triplets of source motion, target motion, and natural-language edit descriptions mined from
 221 AMASS [17], covering body-part adjustments, temporal changes, and style modifications.

4.2 Experimental Design

We train the motion editing model on MotionFix, using source–target motion pairs with their text instructions as supervised signal. Following the evaluation protocol of [1], we report results on 32 randomly sampled batches from the test set, averaged across batches to eliminate evaluation randomness, alongside full test-set results. Generated-to-Source Retrieval results are provided in Appendix E. For motion prediction, the input consists of a source motion sequence paired with a text instruction. For FineFS, we use 30-frame prefixes to predict 40-frame futures, with instructions augmented by a rotation count token (e.g., "double", "triple") to resolve the physical ambiguity inherent in technical takeoffs, where identical trajectories can lead to varied rotational outcomes. For Human3.6M, we use 25-frame prefixes to predict 100-frame futures, with standard action-type tokens (e.g., "eating", "walking") to maintain architectural consistency across datasets.

The scoring reward r_{score} (see Section 3.2.3) is applied only to FineFS, where GOE score annotations are available, achieving 82.13% pairwise accuracy and 90.22% for score margins ≥ 2 . For Human3.6M and MotionFix, we set $\lambda_{\text{score}} = 0$. The reward mechanism is nevertheless general and readily extensible to any dataset with motion quality annotations.

4.3 Baselines and Metrics

Baselines. For motion editing, we compare against MDM [28] and TMED [1] designed by MotionFix. For motion prediction on Human3.6M, we compare against representative VAE-based models [24], [38], [37], [19], [6] and diffusion-based approaches [31], [4], [2], [29], [27], [23]. For FineFS, we select the top-performing Human3.6M baselines: Transfusion [29], CoMusion [27] and TCD [23] to evaluate the precision of ReasonMotion in localized, fine-grained instruction grounding.

Metrics. For motion editing, we report R@k ($\uparrow$), and AvgR ($\downarrow$). For motion prediction, we report ADE, FDE, MMADE, MMFDE, A-MPJPE, and F-MPJPE ($\downarrow$). For FineFS, we additionally report the GOE score ($\uparrow$) to demonstrate the performance gain from reward-guided reasoning. Metric definitions are provided in Appendix C.

Human Evaluation. To complement quantitative metrics, we conducted a formal human evaluation study with eleven figure skating experts, including professional coaches and national-level athletes. Participants assessed the generated motions across four dimensions: (1) **Realism**: experts identify motions indistinguishable from real movement. (2) **Quality Ranking**: experts rank model outputs and reference reconstructed motion by GOE. (3) **Physical Plausibility**: experts rate the coherence of takeoff, flight, and landing phases on a 1-5 Likert scale and assess kinematic consistency of aerial rotations. (4) **Training Utility**: experts evaluate the executability and diagnostic value of AI-proposed corrections. Evaluation details are provided in Appendix D.

5 Results

We evaluate ReasonMotion across three distinct dimensions: fine-grained instruction grounding (Table 1), motion prediction (Table 2 and Table 3), and architectural ablation (FineFS in Table 3 and Human3.6M in Appendix F). Finally, we validate its practical efficacy through an expert human evaluation (Table 4) and qualitative results (Fig 3).

Table 1: Generated-to-Target Retrieval comparison to baselines on the MotionFix benchmark.

Model	Full Test Set				Sampled Batches			
	R@1	R@2	R@3	AvgR	R@1	R@2	R@3	AvgR
GT	64.36	88.75	95.56	1.74	100.0	100.0	100.0	1.00
MDM [28]	0.00	0.30	0.49	476.65	4.03	7.56	10.48	15.55
TMED [1]	18.77	29.64	37.55	56.63	58.96	76.25	82.08	2.71
ReasonMotion	47.24	66.14	73.23	6.50	73.96	80.21	86.46	2.47

Table 2: Comparison to baselines on the Human3.6M dataset for the motion generation task.

Type	Model	ADE ↓	FDE ↓	MMADE ↓	MMFDE ↓
VAE-based	Motron [24]	0.375	0.488	0.509	0.539
	DSF [38]	0.493	0.592	0.550	0.599
	DLow [37]	0.425	0.518	0.495	0.531
	GSPS [19]	0.389	0.496	0.476	0.525
	DivSamp [6]	0.370	0.485	0.475	0.516
DM-based	MotionDiff [31]	0.411	0.509	0.508	0.536
	HumanMAC [4]	0.369	0.480	0.509	0.545
	BeLFusion [2]	0.372	0.474	0.473	0.507
	TransFusion [29]	0.358	0.468	0.506	0.539
	CoMusion [27]	0.350	0.458	0.494	0.506
	TCD (Base) [23]	0.331	0.431	0.482	0.503
	ReasonMotion	0.325	0.423	0.466	0.492

As shown in Table 1, ReasonMotion achieves 73.96% R@1 on the MotionFix benchmark, substantially outperforming strong baseline TMED (58.96%), demonstrating that ReasonMotion effectively maps semantic instructions to localized skeletal adaptations. For the generation task, table 2 confirms this reasoning-driven approach generalizes effectively. ReasonMotion achieves competitive results on Human3.6M, demonstrating that ReasonMotion provides a robust inductive bias for general human kinematics.

On the FineFS dataset (Table 3), ReasonMotion achieves a superior GOE mean of 3.17, significantly outperforming all baselines. While this high score confirms the optimization efficacy of vis-GRPO, we rely on rigorous expert human evaluation to validate that these numerical gains translate to true technical quality and biomechanical plausibility. Interestingly, we observe a “Metric Paradox” where the baseline CoMusion records a marginally lower A-MPJPE (162.89) than our model (171.22), yet produces technically invalid motions (GOE: 0.45), a failure further confirmed by expert human evaluations and qualitative result Figure 3. This indicates that standard Euclidean metrics fail to capture physical integrity, often penalizing minor temporal phase shifts over anatomical collapse. We analyze this paradox in depth in Section 6. Most notably, on the $\text{GOE} > 1$ testing subset, comprising exclusively high-quality jumps, ReasonMotion significantly reduces its A-MPJPE to 155.92. This confirms that the model captures the stable physical logic of elite athletic performance, enabling the high-fidelity reconstruction of successful kinematic chains.

Our ablation study further dissects the contributions of our core modules in Table 3. Results reveal a clear architectural synergy: ST-MoE provides consistent baseline improvements, while vis-GRPO delivers a substantial leap in trajectory accuracy, culminating in the lowest overall error. Notably, while ReasonMotion establishes strong performance on general human kinematics (In Appendix F), its error reduction is far more pronounced on FineFS. This demonstrates that our reasoning-driven pipeline is uniquely effective for the rigorous physical constraints of complex sports motions.

Human Evaluation To validate real-world applicability, we conducted a blind evaluation with professional figure skating coaches and athletes (Table 4). ReasonMotion achieved a 93.3% selection

Table 3: Comparison to baselines on the FineFS dataset for the motion generation task.

Model	A-MPJPE ↓	F-MPJPE ↓	GOE (mean ± std) ↑
GT	0.00	0.00	(real) 0.23 ± 2.95 / (pred) 0.34 ± 2.68[†]
Transfusion [29]	199.83	239.68	0.19 ± 0.12
CoMusion [27]	162.89	187.14	0.45 ± 0.17
TCD (Base) [23]	218.80	236.40	0.51 ± 0.24
TCD + ST-MoE	208.06	212.41	0.49 ± 0.13
TCD + vis-GRPO	175.99	193.81	3.08 ± 0.75
ReasonMotion	171.22	188.27	3.17 ± 0.84
ReasonMotion (GOE > 1)	155.92	172.56	–

[†] GT: (Annotated Labels ± std) / (Reward Model Predictions ± std).

Table 4: **Human Evaluation Results.** Selection rate for visual realism, Top-1 ranking for technical quality (GOE), biomechanical metrics (Rotation Error, Flight, and Landing), and instructional utility.

Model	Realism	Expertise	Biomechanical Plausibility			Utility
	(Select %)	(Top-1 %)	Rot. Err.↓	Flight↑	Land.↑	(Helpful %)
Reference Reconstructed Motion	53.3%	16.2%	$0.35 \pm .71$	$3.40 \pm .97$	3.00 ± 1.18	-
TransFusion [29]	3.3%	1.2%	$1.80 \pm .90$	$1.75 \pm .70$	$1.85 \pm .73$	-
CoMusion [27]	3.3%	0.0%	$1.75 \pm .77$	$1.75 \pm .77$	$1.65 \pm .85$	-
ReasonMotion (Ours)	94.7%	82.5%	$0.33 \pm .43$	$3.85 \pm .85$	$3.85 \pm .91$	82.2%

* '-' denotes not applicable; utility is only evaluated for ReasonMotion outputs[cite: 177].

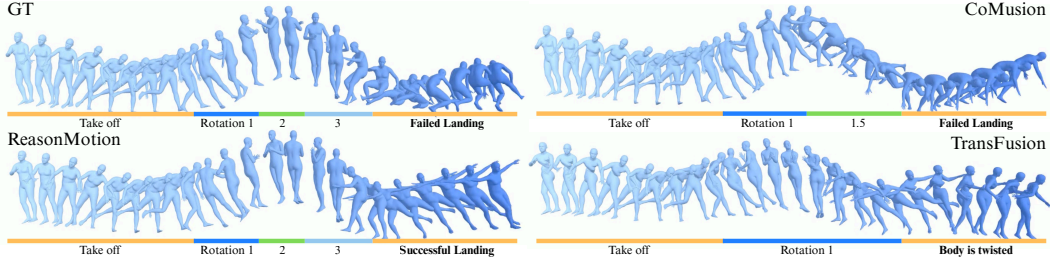


Figure 3: Qualitative Analysis on a 3 Axel jump.

rate for visual realism, outperforming the reference reconstructed motion (50.4%). This counter-intuitive result stems from the model’s denoising capabilities: ReasonMotion filters out jitter in the HybriK reconstructed motion, generating an idealized form perceived as more natural by experts.

Crucially, ReasonMotion was ranked first in technical quality (GOE) in 83.3% of the trials. This expert consensus provides complementary evidence that the reward-guided improvements correspond to plausible skating technique rather than merely exploiting obvious reward-model artifacts. Instead, the model genuinely synthesizes elite-level athletic execution recognized by professionals.

Furthermore, we evaluated biomechanical plausibility across jump phases. ReasonMotion demonstrated superior physical consistency with a minimal rotation error of 0.22, and achieved the highest expert ratings for both flight and landing biomechanics. In contrast, baseline models frequently failed to simulate the required rotations, often collapsing into anatomically implausible postures. Beyond synthesis, experts rated 80.6% of ReasonMotion’s technical corrections as highly valid for training. Notably, among these cases, 100% effectively aided users in visualizing the required improvements, and 87.5% successfully identified the athletes’ biomechanical blind spots. This establishes ReasonMotion as an actionable, biomechanical “what-if” simulator for technical sports.

6 Discussion

Our evaluation reveals two key findings. First, despite underperforming the baseline in MPJPE (Table 3), qualitative inspection reveals the opposite conclusion (Table 4 and Fig. 3): the baseline produces “static” postures from supervised dataset averaging, whereas ReasonMotion discovers a stable kinetic chain consistent with elite athletic form, highlighting a fundamental limitation of MPJPE as a metric for biomechanical quality. Second, through RL refinement, the model progressively shifts from realism toward optimality: early training favors naturalistic motion, while later stages discover elite compensatory strategies, providing a theoretical upper bound for successful landings under suboptimal takeoff conditions (see Appendix I for detailed qualitative analysis).

7 Conclusion

We present ReasonMotion, a framework that transforms motion generation into a strategic sports analysis tool. By combining ST-MoE with vis-GRPO, we move beyond imitation toward optimal kinetic solutions, functioning as a biomechanical “what-if” simulator that lets coaches explore consequences of technical adjustments and paving the way for intelligent athletic consultation.

References

- [1] Nikos Athanasiou, Alpár Cseke, Markos Diomataris, Michael J Black, and Gül Varol. Motionfix: Text-driven 3d human motion editing. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–11, 2024.
- [2] German Barquero, Sergio Escalera, and Cristina Palmero. Belfusion: Latent diffusion for behavior-driven human motion prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2317–2327, 2023.
- [3] Zhi Cen, Huaijin Pi, Sida Peng, Zehong Shen, Minghui Yang, Shuai Zhu, Hujun Bao, and Xiaowei Zhou. Generating human motion in 3d scenes from text descriptions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1855–1866, June 2024.
- [4] Ling-Hao Chen, Jiawei Zhang, Yewen Li, Yiren Pang, Xiaobo Xia, and Tongliang Liu. Humanmac: Masked motion completion for human motion prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9544–9555, 2023.
- [5] Wenxun Dai, Ling-Hao Chen, Jingbo Wang, Jinpeng Liu, Bo Dai, and Yansong Tang. Motionlcm: Real-time controllable motion generation via latent consistency model. In *European Conference on Computer Vision*, pages 390–408. Springer, 2024.
- [6] Lingwei Dang, Yongwei Nie, Chengjiang Long, Qing Zhang, and Guiqing Li. Diverse human motion prediction via gumbel-softmax sampling from an auxiliary space. In *Proceedings of the 30th ACM international conference on multimedia*, pages 5162–5171, 2022.
- [7] Markos Diomataris, Nikos Athanasiou, Omid Taheri, Xi Wang, Otmar Hilliges, and Michael J. Black. Wandr: Intention-guided human motion generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 927–936, June 2024.
- [8] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3d human motions from text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5152–5161, June 2022.
- [9] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-rl: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [10] Wen Guo, Yuming Du, Xi Shen, Vincent Lepetit, Xavier Alameda-Pineda, and Francesc Moreno-Noguer. Back to mlp: A simple baseline for human motion prediction. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 4809–4819, 2023.
- [11] Ziyang Guo, Zeyu Hu, Na Zhao, and De Wen Soh. Motionlab: Unified human motion generation and editing via the motion-condition-motion paradigm. *International Conference on Computer Vision (ICCV)*, 2025.
- [12] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu. Human3. 6m: Large scale datasets and predictive methods for 3d human sensing in natural environments. *IEEE transactions on pattern analysis and machine intelligence*, 36(7):1325–1339, 2013.
- [13] Jiefeng Li, Chao Xu, Zhicun Chen, Siyuan Bian, Lixin Yang, and Cewu Lu. Hybrik: A hybrid analytical-neural inverse kinematics solution for 3d human pose and shape estimation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3383–3393, 2021.
- [14] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- [15] Shaowei Liu, Yang Zhou, Jimei Yang, Saurabh Gupta, and Shenlong Wang. Contactgen: Generative contact modeling for grasp generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20609–20620, 2023.
- [16] Wenhan Ma, Hailin Zhang, Liang Zhao, Yifan Song, Yudong Wang, Zhifang Sui, and Fuli Luo. Stabilizing moe reinforcement learning by aligning training and inference routers, 2025.
- [17] Naureen Mahmood, Nima Ghorbani, Nikolaus F Troje, Gerard Pons-Moll, and Michael J Black. Amass: Archive of motion capture as surface shapes. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5442–5451, 2019.

- [18] Gabriel Maldonado, Armin Danesh Pazho, Ghazal Alinezhad Noghre, Vinit Katariya, and Hamed Tabkhi. Moclip motion-aware fine-tuning and distillation of clip for human motion generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR) Workshops*, pages 2931–2941, June 2025.
- [19] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. Generating smooth pose sequences for diverse human motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 13309–13318, 2021.
- [20] David McAllister, Songwei Ge, Brent Yi, Chung Min Kim, Ethan Weber, Hongsuk Choi, Haiwen Feng, and Angjoo Kanazawa. Flow matching policy gradients, 2025.
- [21] Liangdong Qiu, Chengxing Yu, Yanran Li, Zhao Wang, Haibin Huang, Chongyang Ma, Di Zhang, Pengfei Wan, and Xiaoguang Han. Vimo: Generating motions from casual videos. *arXiv preprint arXiv:2408.06614*, 2024.
- [22] Jose Ribeiro-Gomes, Tianhui Cai, Zoltán A Milacski, Chen Wu, Aayush Prakash, Shingo Takagi, Amaury Aubel, Daeil Kim, Alexandre Bernardino, and Fernando De La Torre. Motiongpt: Human motion synthesis with improved diversity and realism via gpt-3 prompting. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 5070–5080, 2024.
- [23] Saeed Saadatnejad, Ali Rasekh, Mohammadreza Mofayez, Yasamin Medghalchi, Sara Rajabzadeh, Taylor Mordan, and Alexandre Alahi. A generic diffusion-based approach for 3d human pose prediction in the wild. In *International Conference on Robotics and Automation (ICRA)*, 2023.
- [24] Tim Salzmann, Marco Pavone, and Markus Ryll. Motron: Multimodal probabilistic human motion forecasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6457–6466, 2022.
- [25] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [26] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- [27] Jiarui Sun and Girish Chowdhary. Comusion: Towards consistent stochastic human motion prediction via motion diffusion. In *European conference on computer vision*, pages 18–36. Springer, 2024.
- [28] Guy Tevet, Sigal Raab, Brian Gordon, Yonatan Shafir, Daniel Cohen-Or, and Amit H. Bermano. Human motion diffusion model, 2022.
- [29] Sibotian, Minghui Zheng, and Xiao Liang. Transfusion: A practical and effective transformer-based diffusion model for 3d human motion prediction. *IEEE Robotics and Automation Letters*, 9(7):6232–6239, 2024.
- [30] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in neural information processing systems*, 33:5776–5788, 2020.
- [31] Dong Wei, Huaijiang Sun, Bin Li, Jianfeng Lu, Weiqing Li, Xiaoning Sun, and Shengxiang Hu. Human joint kinematics diffusion-refinement for stochastic motion prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 6110–6118, 2023.
- [32] Qi Wu, Yubo Zhao, Yifan Wang, Xinhang Liu, Yu-Wing Tai, and Chi-Keung Tang. Motion-agent: A conversational framework for human motion generation with LLMs. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [33] Yi Xu and Yun Fu. Sports-traj: A unified trajectory generation model for multi-agent movement in sports. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [34] Hongdi Yang, Chengyang Li, Zhenxuan Wu, Gaozheng Li, Jingya Wang, Jingyi Yu, Zhuo Su, and Lan Xu. Simgdiff: Soccer motion generation using diffusion probabilistic models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11807–11817, 2025.
- [35] Huili Huang Lijing Mao Yang Zhou Yanli Ji, Lingfeng Ye and Lingling Gao. Localization-assisted uncertainty score disentanglement network for action quality assessment. In *ACM MM*, pages 1–10, 2023.

- 415 [36] Wei-Hsin Yeh, Yu-An Su, Chih-Ning Chen, Yi-Hsueh Lin, Calvin Ku, Wenhsin Chiu, Min-Chun Hu, and
 416 Lun-Wei Ku. Coachme: Decoding sport elements with a reference-based coaching instruction generation
 417 model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*
 418 *(Volume 1: Long Papers)*, pages 29126–29151, 2025.
- 419 [37] Ye Yuan and Kris Kitani. Dlow: Diversifying latent flows for diverse human motion prediction. In
 420 *European Conference on Computer Vision*, pages 346–364. Springer, 2020.
- 421 [38] Ye Yuan and Kris M Kitani. Diverse trajectory forecasting with determinantal point processes. In
 422 *International Conference on Learning Representations*, 2020.
- 423 [39] Ling-An Zeng, Guohong Huang, Gaojie Wu, and Wei-Shi Zheng. Light-t2m: A lightweight and fast model
 424 for text-to-motion generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39,
 425 pages 9797–9805, 2025.
- 426 [40] Jianrong Zhang, Yangsong Zhang, Xiaodong Cun, Yong Zhang, Hongwei Zhao, Hongtao Lu, Xi Shen,
 427 and Ying Shan. Generating human motion from textual descriptions with discrete representations. In
 428 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14730–14740,
 429 2023.
- 430 [41] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu.
 431 Motiondiffuse: Text-driven human motion generation with diffusion model. *IEEE transactions on pattern*
 432 *analysis and machine intelligence*, 46(6):4115–4128, 2024.
- 433 [42] Yaqi Zhang, Di Huang, Bin Liu, Shixiang Tang, Yan Lu, Lu Chen, Lei Bai, Qi Chu, Nenghai Yu, and
 434 Wanli Ouyang. Motiongpt: Finetuned llms are general-purpose motion generators. In *Proceedings of the*
 435 *AAAI Conference on Artificial Intelligence*, volume 38, pages 7368–7376, 2024.
- 436 [43] Zeyu Zhang, Yiran Wang, Wei Mao, Danning Li, Rui Zhao, Biao Wu, Zirui Song, Bohan Zhuang, Ian Reid,
 437 and Richard Hartley. Motion anything: Any to motion generation. *arXiv preprint arXiv:2503.06955*, 2025.
- 438 [44] Zixiang Zhou, Yu Wan, and Baoyuan Wang. Avatargpt: All-in-one framework for motion understanding
 439 planning generation and beyond. In *Proceedings of the IEEE/CVF Conference on Computer Vision and*
 440 *Pattern Recognition*, pages 1357–1366, 2024.
- 441 [45] Wentao Zhu, Xiaoxuan Ma, Dongwoo Ro, Hai Ci, Jinlu Zhang, Jiaxin Shi, Feng Gao, Qi Tian, and
 442 Yizhou Wang. Human motion generation: A survey. *IEEE Transactions on Pattern Analysis and Machine*
 443 *Intelligence*, 46(4):2430–2449, 2023.

444 A vis-GRPO Algorithm

445 In this appendix, we provide the detailed training procedure for the Motion Prediction task using
446 vis-GRPO.

Algorithm 1 Training: Motion Predictor (vis-GRPO Stage)

Require: SFT-trained Predictor M_P^{SFT}
Require: Instruction token S , Steps T , Schedules $\{\alpha_t\}$

```

0: Initialize Policies:  $\pi_\theta, \pi_{\theta_{\text{old}}}, \pi_{\text{ref}} \leftarrow M_P^{\text{SFT}}$ 
0: Split  $x_0 \rightarrow (x_{\text{past}}, x_{\text{future}})$ 
0:  $t \sim \mathcal{U}\{1, \dots, T\}, \varepsilon \sim \mathcal{N}(0, I)$ 
0:  $x_t \leftarrow \sqrt{\alpha_t} \cdot x_{\text{future}} + \sqrt{1 - \alpha_t} \cdot \varepsilon$ 
0:  $z_t \leftarrow \text{TIMEEMBED}(t), e_S \leftarrow \text{TOKENEMBED}(S)$ 
0:  $h \leftarrow \text{ENCODECONDITION}(x_{\text{past}}, z_t, e_S)$ 
0: for  $i = 1$  to  $G$  do {Sample  $G$  candidates}
0:    $\hat{\varepsilon}_i \leftarrow M_P(x_t, h)$  {Denoise using policy  $\pi_\theta$ }
0:    $\hat{x}_{\text{future}}^{(i)} \leftarrow \text{REVERSEDIFFUSION}(x_t, \hat{\varepsilon}_i, t)$ 
0:    $r_i \leftarrow \text{REWARDMODEL}(\hat{x}_{\text{future}}^{(i)}, x_{\text{future}}, S)$ 
0: end for
0: Compute  $\pi(o_i | q)$  for  $\pi_\theta, \pi_{\theta_{\text{old}}}, \pi_{\text{ref}}$  over  $\{\hat{x}_{\text{future}}^{(i)}\}_{i=1}^G$ 
0: Compute  $A_i = (r_i - \text{mean}(\mathbf{r})) / (\text{std}(\mathbf{r}) + \epsilon_A)$ 
0:  $\mathcal{L} \leftarrow -J_{\text{vis-GRPO}}(\theta)$ 
0: UPDATEPOLICY( $\pi_\theta, \mathcal{L}$ ) {Update  $\pi_\theta$  weights per batch}
0: // At the end of each epoch
0:  $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta = 0$ 

```

447 **Rollout Routing Replay (R3) for GRPO.** Following the R3 framework [16], we record the expert
448 assignment $\mathcal{I}_t^{\text{infer}}$ at each denoising step t during the rollout pass and replay it during the training
449 pass. The router logits $s^{\text{train}} = W_r x^{\text{train}}$ are still computed from the current parameters to preserve
450 gradient flow, but expert selection is fixed to $\mathcal{I}_t^{\text{infer}}$. Gate weights are recomputed via a masked
451 softmax restricted to the recorded experts:

$$g_i^{\text{R3}} = \frac{\exp(s_i^{\text{train}})}{\sum_{j \in \mathcal{I}_t^{\text{infer}}} \exp(s_j^{\text{train}})}, \quad i \in \mathcal{I}_t^{\text{infer}} \quad (13)$$

$$\text{MoE-FFN}^{\text{R3}}(x^{\text{train}}) = \sum_{i \in \mathcal{I}_t^{\text{infer}}} g_i^{\text{R3}} E_i(x^{\text{train}}) \quad (14)$$

453 This formulation ensures: (1) expert assignment is aligned between rollout and training passes, so the
454 policy gradient is computed on the same computational path that produced the trajectory; and (2) the
455 gradient with respect to router weights W_r is preserved through the masked softmax, allowing the
456 router to continue learning.

457 Since the denoising chain consists of T sequential forward passes with distinct inputs x_t , R3 records
458 and replays routing independently at each step, producing a routing cache $\{\mathcal{I}_t^{\text{infer}}\}_{t=T-1}^0$:

$$\text{Rollout: } x_t \xrightarrow{\mathcal{I}_t^{\text{infer}}} x_{t-1}, \quad \text{Training: } x_t \xrightarrow{\mathcal{I}_t^{\text{infer}}} \nabla_\theta \mathcal{L}_t, \quad \forall t \in \{T-1, \dots, 0\}. \quad (15)$$

459 B Reward Model Details

460 In this appendix, we detail the formula for each individual reward functions.

461 B.1 Scoring Reward

462 The Scoring Reward Model is implemented as a two-stage network. The first stage is a Human Pose
463 Perception (HPP) module adapted from CoachMe [36], which encodes the spatio-temporal structure
464 of motion sequences into compact skeletal features. The second stage is a feedforward regression
465 head that maps pooled features to a scalar score:

$$r_{\text{score}}(x) = \text{MLP}(\text{HPP}(x)) \quad (16)$$

$$= \mathbf{w}_2^\top \sigma(\mathbf{w}_1^\top \text{Pool}(\text{HPP}(x)) + b_1) + b_2, \quad (17)$$

466 where $x \in \mathbb{R}^{C \times T \times V}$ is the input motion (channels $\times$ time $\times$ joints), $\text{HPP}(\cdot)$ extracts skeletal features,
467 and $\text{Pool}(\cdot)$ denotes spatio-temporal adaptive average pooling. The head is trained via mean squared
468 error against ground-truth GOE labels.

469 B.2 Motion Similarity Reward

470 The motion similarity reward penalizes L2 deviation between the generated motion $\hat{x}$ and the ground-
471 truth motion x , both flattened into vectors in $\mathbb{R}^D$ where $D = K \times L \times 3$:

$$r_{\text{similarity}} = -\|\hat{x} - x\|_2. \quad (18)$$

472 B.3 Motion Quality Reward

473 The motion quality reward encourages physical plausibility via two terms. The smoothness reward
474 penalizes rapid fluctuations in velocity and acceleration:

$$r_{\text{smooth}} = -(\|\Delta^2 \hat{x}\|_2 + \|\Delta^3 \hat{x}\|_2), \quad (19)$$

475 where $\Delta^2 \hat{x}$ and $\Delta^3 \hat{x}$ denote discrete second- and third-order finite differences. The bone consistency
476 reward penalizes temporal variation in distances between connected joint pairs $(j_1, j_2) \in \mathcal{B}$:

$$r_{\text{bone}} = -\text{Std}_t(\|\hat{x}_{j_1, t} - \hat{x}_{j_2, t}\|_2). \quad (20)$$

477 The final quality reward is their weighted sum:

$$r_{\text{quality}} = \lambda_{\text{smooth}} \cdot r_{\text{smooth}} + \lambda_{\text{bone}} \cdot r_{\text{bone}}. \quad (21)$$

C Evaluation Metrics Definitions

We provide detailed definitions of the evaluation metrics used in the main paper.

R Precision (R@k). To assess whether a generated motion sequence conveys the *intended semantic edit*, we employ R Precision (Recall@k). Let $\{\hat{\mathbf{X}}_i\}_{i=1}^N$ and $\{\mathbf{X}_i\}_{i=1}^N$ denote N generated sequences and their corresponding ground-truths. Both are embedded with a fixed, pretrained motion encoder (512-D), yielding feature vectors $\mathbf{z}_i^{\text{gen}}$ and $\mathbf{z}_i^{\text{gt}} \in \mathbb{R}^{512}$. Cosine similarity is computed as:

$$s_{ij} = \frac{\mathbf{z}_i^{\text{gen}} \cdot \mathbf{z}_j^{\text{gt}}}{\|\mathbf{z}_i^{\text{gen}}\|_2 \|\mathbf{z}_j^{\text{gt}}\|_2}, \quad 1 \leq i, j \leq N. \quad (22)$$

For each generated sequence i , we calculate the rank r_i of the ground-truth among all N candidates by descending similarity. R Precision is the recall of correct matches within the top k :

$$\text{R@}k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[r_i \leq k], \quad k \in \{1, 2, 3\}. \quad (23)$$

In addition, we report the **Average Rank (AvgR)**.

MPJPE (Mean Per Joint Position Error). MPJPE is our primary evaluation metric for position accuracy. For a given frame t , MPJPE is defined as the average Euclidean distance between the predicted joint positions $\hat{x}_{j,t}$ and the ground-truth positions $x_{j,t}$:

$$\text{MPJPE}(t) = \frac{1}{J} \sum_{j=1}^J \|\hat{x}_{j,t} - x_{j,t}\|_2, \quad (24)$$

where J is the number of joints. We report two variants:

- **A-MPJPE:** Mean MPJPE averaged over all predicted future frames in the generated sequence.
- **F-MPJPE:** MPJPE computed only in the final predicted frame.

ADE (Average Displacement Error). For human motion prediction, we treat each predicted future pose as a vector $\hat{\mathbf{y}}_t \in \mathbb{R}^{3J}$. Given M generated hypotheses, ADE is computed using the best matching hypothesis:

$$\text{ADE} = \min_{m=1,\dots,M} \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{y}}_t^{(m)} - \mathbf{y}_t\|_2. \quad (25)$$

FDE (Final Displacement Error). FDE measures the final-frame pose error using the best hypothesis:

$$\text{FDE} = \min_{m=1,\dots,M} \|\hat{\mathbf{y}}_T^{(m)} - \mathbf{y}_T\|_2. \quad (26)$$

MMADE. For multi-modal evaluation, each test sample may have a set of plausible ground-truth futures $\mathcal{Y}_i$. MMADE averages the best-hypothesis ADE over all valid ground-truth futures:

$$\text{MMADE}_i = \frac{1}{|\mathcal{Y}_i|} \sum_{\mathbf{y} \in \mathcal{Y}_i} \min_{m=1,\dots,M} \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{y}}_t^{(m)} - \mathbf{y}_t\|_2. \quad (27)$$

MMFDE. MMFDE averages the best-hypothesis final-frame error over all plausible ground-truth futures:

$$\text{MMFDE}_i = \frac{1}{|\mathcal{Y}_i|} \sum_{\mathbf{y} \in \mathcal{Y}_i} \min_{m=1,\dots,M} \|\hat{\mathbf{y}}_T^{(m)} - \mathbf{y}_T\|_2. \quad (28)$$

where T is the prediction horizon, J is the number of joints, M is the number of hypotheses generated, $\hat{\mathbf{y}}_t^{(m)}$ denotes the m -th predicted pose vector at time t , $\mathbf{y}_t$ is the ground-truth pose vector, and $\|\cdot\|_2$ is the Euclidean norm. For multi-modal evaluation, $\mathcal{Y}_i$ denotes the set of plausible ground-truth futures for test sample i .

D Human Evaluation Details

In this appendix, we provide the comprehensive details of our human evaluation experiment, including participant, stimuli selection, interface implementation, specific evaluation phases, and statistical analysis procedures.

D.1 Participant Information

Recruitment Criteria. To ensure high-quality and domain-specific feedback, we established rigorous qualification criteria for our evaluators. Participants are classified as “figure skating experts” if they meet at least one of the following criteria:

- Certified professional figure skating coaches actively training competitive athletes.
- Active competitive/elite figure skating athletes with specialized experience in multi-rotation jumps (double and triple jumps).

Evaluator Cohort. We have conducted pilot studies with 11 high-fidelity expert evaluators who completed the survey with extreme diligence: four professional coaches and seven competitive international level figure skating athletes.

Table 5: **Evaluator Demographics.** Our study cohort ($n = 11$) consists of elite figure skating professionals with extensive competitive and instructional experience.

Role	Level / Experience	Count	Technical Milestone
Professional Coach	Int’l/National (Avg. 6.8 yrs)	4	Coached Int’l/National Medalists
Elite Athlete	National Team (Avg. 10.3 yrs)	7	Confirmed Triple-jump proficiency

Individual Evaluator Background. To ensure the highest standard of technical assessment, our cohort comprises elite members of the figure skating community. The coaches possess an average of 6.8 years of professional instructional experience, with several having successfully trained athletes for international ISU competitions. The athlete group consists of active national-level competitors with an average of 10.3 years of experience, all of whom have confirmed technical proficiency in triple-jump execution (Table 6).

Table 6: **Detailed Expertise Profiles of Evaluators** ($n = 11$).

ID	Role	Experience	Competitive / Coaching Milestone
C-1	Coach	10 Years	Trained athletes for International Competitions
C-2	Coach	8 Years	Trained athletes for National Competitions
C-3	Coach	5 Years	Former National Athlete; National-level Coach
C-4	Coach	4 Years	Former National Athlete; National-level Coach
A-1	Athlete	15 Years	National Team Member; Triple-jump proficient
A-2	Athlete	16 Years	National Team Member; Triple-jump proficient
A-3	Athlete	5 Years	International competitor; Double-jump proficient
A-4	Athlete	5 Years	International competitor; Double-jump proficient
A-5	Athlete	7 Years	International competitor; Triple-jump proficient
A-6	Athlete	7 Years	National Team Member; Triple-jump proficient
A-7	Athlete	9 Years	National Team Member; Triple-jump proficient

Ethics and Consent. Participation in this study is entirely voluntary and conducted anonymously. The platform collects the names and roles of the evaluators solely to prevent duplicate entries and to enable session-saving (progress can be saved mid-survey and resumed by entering the same identity, which is crucial for busy professional coaches and athletes). Evaluators are explicitly informed that their identity will not be linked to their responses.

D.2 Stimuli and Setup

Case Selection (Pools). Our evaluation stimuli are sampled from the testing set of the proposed dataset. Specifically, we focus on double and triple jumps because:

1. Multi-rotational jumps are highly representative of technical difficulty in figure skating.
2. These sequences exhibit high-fidelity motion capture quality, ensuring that artifacts from data collection do not bias the evaluation of generative performance.

The specific pool consists of 14 jump cases divided into four categories:

- **2-Turns Negative:** Unsuccessful double jumps (e.g., 2A_0276, 2T_0006).
- **2-Turns Positive:** Successful double jumps (e.g., 2A_0394, 2A_0468).
- **3-Turns Negative:** Unsuccessful triple jumps (e.g., 3A_0057, 3A_0477, 3F_0086, 3Lo_0152, 3Lz_0140).
- **3-Turns Positive:** Successful triple jumps (e.g., 3A_0160, 3A_0402, 3A_0421, 3F_0095, 3Lo_0039, 3Lz_0239).

Models Compared. We compare four distinct conditions:

1. **Ground Truth (GT):** Raw/reconstructed human expert motion.
2. **ReasonMotion_rl** (Our proposed model).
3. **CoMusion** [27] (Baseline).
4. **TransFusion** [29] (Baseline).

Presentation Format. All animations are presented as high-fidelity, loop-rendered 3D human body sequences using the SMPL body model overlay. To allow experts to scrutinize subtle joint placements and takeoff mechanics. All model identifiers are completely blinded and randomized under placeholder labels (e.g., “Video 1” to “Video 4”).

Randomization and Counterbalancing. To avoid order bias:

- The section orders (Realism, GOE Ranking, Training Utility, and Physicality) and trial orders are randomized.
- The relative column positions of the videos within each grid are randomized per trial.
- Shuffling is performed deterministically using a user-specific stable hashing seed generated from their Name and Role, ensuring a uniform distribution across conditions while maintaining consistent state across browser refreshes.

D.3 Phase-Specific Instructions and Interfaces

Phase 1: Realism Test(15 Questions, 60 Videos Evaluated).

- **Instruction Prompt:** “Which of these look real? Please select all videos that appear realistic (you can select more than one !!!!). Note: Due to potential errors in motion capture and human shape rendering (e.g., twisting, body shape distortion, hunchback, etc.), please use the realism of the movement posture and jump as the primary basis for judgment.”
- **Interface Layout:** A grid of four videos corresponding to the four models.
- **Evaluation Format:** Multiple-choice checkbox. Evaluators can select any number of videos (from 1 to 4) they deem realistic.

Phase 2: Quality (GOE) Ranking (8 Questions, 32 Videos Evaluated).

- **Instruction Prompt:** “GOE Execution Quality Ranking. Please rank the following videos by execution quality (GOE) from high to low.”
- **Interface Layout:** A grid of four blinded videos with individual dropdown menus.
- **Evaluation Format:** Forced-unique ranking from 1 (Best) to 4 (Worst). The frontend ensures that evaluators cannot assign the same rank to multiple videos in a single trial.

576 **Phase 3: Training Utility (4–24 Questions, 8 Videos Evaluated).** This phase specifically targets
577 corrective generation, using unsuccessful jumps (Negative pools) as input.

- 578 • **Instruction Prompt:** “Does watching the corrected motion video help your training? In
579 this stage, we use the take-off phase of the jump on the left as input to generate a video of a
580 jump that theoretically has a higher GOE in the subsequent ‘air spin’ and ‘landing.’ If you
581 were the skater on the left and performed a poor jump, could the video on the right serve as
582 a demonstration to help you understand how to improve? The right video is a vertical stack
583 of three videos: a real motion (grey) and two corrected motions, one demonstrating a lighter
584 correction (light blue), and one demonstrating a stronger correction (dark blue).”
- 585 • **Interface Layout:** A split-screen showing the original failed jump on the left and the
586 overlay/corrected stack on the right.
- 587 • **Questions and Logic:**
 - 588 1. *Primary Question:* “Is this corrected video helpful for your training?” (Binary choice:
589 Yes / No).
 - 590 2. *Conditional Sub-Questions* (rendered only if the primary answer is Yes):
 - 591 – **Executability:** “How high is the ‘executability’ of the physical adjustments pre-
592 sented in real training?” (5-point Likert scale, anchors: 1: Very Low, 2: Low, 3:
593 Fair, 4: High, 5: Very High).
 - 594 – **Blind Spot Detection:** “Does the video point out blind spots?” (Yes / No).
 - 595 – **Problem Understanding:** “Better understanding of issue?” (Yes / No).
 - 596 – **Improvement Visualization:** “Help you visualize improvement?” (Yes / No).
 - 597 – **Inspiration Source:** “Main inspiration?” (Multi-select checkboxes: Arms,
598 Landing, Core).

599 **Phase 4: Physical Plausibility (32 Questions, 8 Videos Evaluated).** To rigorously assess kine-
600 matic violations, each model’s output is evaluated independently.

- 601 • **Instruction Prompt:** “Is the video physically plausible? Please rate the plausibility of the
602 movement.”
- 603 • **Questions and Scale Anchors:**
 - 604 1. *Air Rotations:* “How many air rotations did you observe in the video?” (Discretized
605 selection: 0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0).
 - 606 2. *Takeoff Phase:* “Takeoff: Is the movement physically plausible and coherent? (e.g.,
607 is the power generation natural? Are there any takeoff from a complete standstill or
608 unrealistic pulling of the center of gravity?)”
 - 609 3. *Flight Phase:* “Flight: Is the movement physically plausible and coherent?”
 - 610 4. *Landing Phase:* “Landing: Is the movement physically plausible and coherent?”

611 All three sub-phases are rated using a 5-point Likert scale (anchors: 1: Very Poor, 2: Poor,
612 3: Fair, 4: Good, 5: Excellent).

613 **D.4 Measurement and Aggregation**

614 For each criterion, we aggregate the responses from our professional evaluators as follows.

615 **Realism** is aggregated as the selection rate, i.e., the percentage of trials in which evaluators selected a
616 given model’s output as the most visually realistic motion.

617 **Expertise (GOE)** is reported as the Top-1 rate, i.e., the percentage of trials in which a model’s output
618 was ranked first in technical quality by the evaluators.

619 **Biomechanical Plausibility** is measured along three dimensions: (1) rotation count error, computed
620 as the absolute difference between the evaluator-counted rotations and the ground truth rotation count;
621 (2) flight phase rating, reported as the mean Likert score (1–5) across all evaluators; and (3) landing
622 phase rating, similarly reported as the mean Likert score (1–5).

623 **Instructional Utility** is aggregated as the percentage of evaluators who indicated that the generated
624 motion was helpful as a coaching demonstration. This metric is only applicable to ReasonMotion
625 outputs, as baseline models do not support instruction-based motion generation.

D.5 Results

Across 10 expert evaluators, we collected a total of 600 video-level realism judgments in Phase 1 (15 trials $\times$ 4 videos $\times$ 10 evaluators), 320 video-level GOE rankings in Phase 2 (8 trials $\times$ 4 videos $\times$ 10 evaluators), and 80 case-level responses each in Phase 3 and Phase 4 (8 cases $\times$ 10 evaluators).

Table 7: Realism & Execution Quality (Phase 1 & 2). Asterisks (*) indicate methods statistically significantly lower ($p < 0.05$) than ReasonMotion. Our model is nearly indistinguishable from Ground Truth.

Model	Realism (%) $\uparrow$	GOE Top-1 (%) $\uparrow$	Mean Rank $\downarrow$
Ref. Reconstructed model	53.3*	16.2	2.08
ReasonMotion (Ours)	94.7	82.5	1.18
CoMusion	3.3*	0.0	3.55*
TransFusion	3.3*	1.2	3.20*

Table 8: Instructional Utility (ReasonMotion only) (Phase 3). The data confirms a clear split: athletes find light corrections easier to execute and focus on landing, whereas coaches find strong corrections more helpful for identifying core biomechanical issues.

Correction Type	Helpful (%) $\uparrow$	Executability (1–5) $\uparrow$	Blind Spot (%) $\uparrow$	Prob. Und. (%) $\uparrow$
Light (Athletes)	79.2%	3.84 ± 0.81	78.9%	78.9%
Strong (Coaches)	87.5%	3.29 ± 0.59	100.0%	92.9%

Table 9: Inspiration Source Distribution (ReasonMotion only) (Phase 3). Athletes primarily focus on landing adjustments under light correction, while coaches shift attention to core biomechanics under strong correction.

Correction Type	Arms (%)	Core (%)	Landing (%)
Light (Athletes)	15.8%	15.8%	68.4%
Strong (Coaches)	14.3%	78.6%	7.1%

Table 10: Biomechanical Plausibility (Phase 4). Likert scores (1–5) reported as mean $\pm$ std. ReasonMotion scores higher than Ground Truth in several biomechanical categories, likely because our model synthesizes “idealized” motion based on instructions.

Model	Rot. Error $\downarrow$	Takeoff $\uparrow$	Flight $\uparrow$	Landing $\uparrow$	Overall Avg $\uparrow$
Reference Reconstructed Motion	0.35 ± 0.71	3.35 ± 1.11	3.40 ± 0.97	3.00 ± 1.18	3.25
ReasonMotion (Ours)	0.33 ± 0.43	3.90 ± 0.94	3.85 ± 0.85	3.85 ± 0.91	3.87
CoMusion	2.05 ± 0.63	2.85 ± 1.19	1.75 ± 0.77	1.65 ± 0.85	2.08
TransFusion	1.80 ± 0.90	2.60 ± 1.16	1.75 ± 0.70	1.85 ± 0.73	2.07

D.6 Qualitative Feedback and User Impressions

Beyond numerical ratings, we collected open-ended feedback to assess the model’s potential for real-world integration into training workflows. The experts highlighted several key advantages:

- **Novelty and Instructional Value:** One senior coach (C-2) noted that the model’s approach to motion adaptation is “*highly novel and promises to be exceptionally helpful for day-to-day technical training.*” This underscores the model’s role as a strategic consultant rather than just a visualization tool.
- **Precision in Error Diagnosis:** An elite athlete (A-2) emphasized the system’s accuracy, stating it was “*very cool to see a system that can effectively identify specific jumps that need improvement*” and expressed a strong willingness to use such a tool in actual practice.

- **Simulation and Prototyping:** Another athlete (A-6) remarked on the practical utility of the framework, finding it particularly useful for “*simulating potential jump strategies*,” allowing athletes to visualize successful outcomes before attempting them physically.

These comments suggest that ReasonMotion provides a unique “coaching sandbox” that aligns with the professional cognitive models of elite skaters.

E Additional Motion Editing results

Here we provide the result on full testing set for motion editing on Generated-to-Source retrieval.

Table 11: Generated-to-Source Retrieval comparison to baselines on the MotionFix benchmark. *Full set*: entire test set. *Sampled*: averaged over 32 randomly sampled batches to eliminate evaluation randomness.

Model	Full Test Set				Sampled Batches			
	R@1	R@2	R@3	AvgR	R@1	R@2	R@3	AvgR
GT	20.83	33.66	40.47	.13	74.01	84.52	89.91	2.03
MDM [28]	0.10	0.10	0.10	486.26	2.62	6.15	9.38	15.88
TMED [1]	29.84	44.47	52.77	31.42	71.46	82.92	88.75	1.96
ReasonMotion	55.21	70.83	76.04	12.02	55.21	70.83	76.04	3.82

F Additional Ablation study

Table 12: **Ablation study of ReasonMotion on Human3.6M.** We evaluate the contribution of ST-MoE and vis-GRPO measured by A-MPJPE and F-MPJPE ($\downarrow$).

Model	A-MPJPE $\downarrow$	F-MPJPE $\downarrow$
TCD (Base)	112.88	146.40
TCD + ST-MoE	110.62	128.32
TCD + vis-GRPO	101.60	130.64
ReasonMotion (Full)	99.66	128.01

G Implementation Details

G.1 Hyperparameters Configuration

This section provides the hyperparameters and configuration settings used to train ReasonMotion.

Table 13: **Implementation Hyperparameters.** Full configuration for all training stages on FineFS and Human3.6M (H36M).

<i>(a) Diffusion Model (TCD Backbone)</i>			<i>(b) Training Configuration</i>		
Parameter	FineFS	H36M	Parameter	FineFS	H36M
Layers	12	12	Epochs	50	50
Joints	24	17	Batch Size	16	16
Channels	64	64	Learning Rate	0.001	2×10^{-4}
Attention Heads	8	8	Optimizer	AdamW	Adam
Time Embedding Dim	128	128	Weight Decay	0.01	1×10^{-6}
Feature Embedding Dim	16	16	Gradient Clipping	1.0	1.0
Beta Start	0.0001	1×10^{-4}	LR Schedule	Constant	2000-step warmup + Cosine Annealing
Beta End	0.5	0.5	Input Frames	30	25
Diffusion Steps (T)	50	50	Output Frames	40	100
Noise Schedule	cosine	cosine	Frame Skip Rate	1	1
			Downsampling Rate	1	1

<i>(c) vis-GRPO Settings</i>			<i>(d) Reward Configuration</i>		
Parameter	FineFS	H36M	Parameter	FineFS	H36M
Policy LR	5×10^{-6}	3×10^{-6}	λ_{score}	0.4	0
Clip Ratio (ϵ)	0.2	0.1	$\lambda_{\text{similarity}}$	0.4	0.7
KL Coef. (β)	1×10^{-5}	1×10^{-2}	λ_{quality}	0.2	0.3
Max Gradient Norm	1.0	1.0	Scaling Method	Min-Max	NULL
Group Size (G)	16	8	Temperature	1.0	–
Old Policy Update	Every epoch	Every 5 batches	<i>r_{quality} breakdown (FineFS only):</i>		
Ref Policy Update	Every 50 samples	Fixed SFT ckpt	Velocity Smoothness	0.25	–
Gaussian Std (σ)	0.03 m	0.015 m	Acceleration Smoothness	0.25	–
			Bone Length Consistency	0.50	–

G.2 Computational Resources

All experiments were conducted on a single NVIDIA RTX 4090 GPU (24GB VRAM). We report the approximate training time for each experimental configuration in Table 14.

Table 14: Computational cost of each experimental run.

Dataset	Stage	Training Time (hrs)	GPU Hours
Human3.6M	Pretraining	12	12
Human3.6M	RL (vis-GRPO)	24	24
HumanFineFS	Pretraining	5	5
HumanFineFS	RL (vis-GRPO)	24	24
MotionFix	Training	30	30
Total		95	95

The total compute across all reported experiments is estimated at approximately 95 GPU hours. Note that this does not account for preliminary or ablation experiments conducted during the development of the method, which required additional compute beyond what is reported here.

G.3 Numerical Stability

To ensure stable training during the RL phase, we implement the following measures:

- **Gradient Clipping:** Gradients are clipped to a maximum norm of 1.0.
- **Value Clamping:** Tensor values are clamped to $[-100, 100]$ to prevent explosions.
- **NaN/Inf Detection:** Automatic detection replaces invalid values with zero or safe constants.
- **Small Epsilon:** A constant $\epsilon_A = 1 \times 10^{-6}$ is added to denominators (, in advantage normalization) to prevent division by zero.

H Training Dynamics

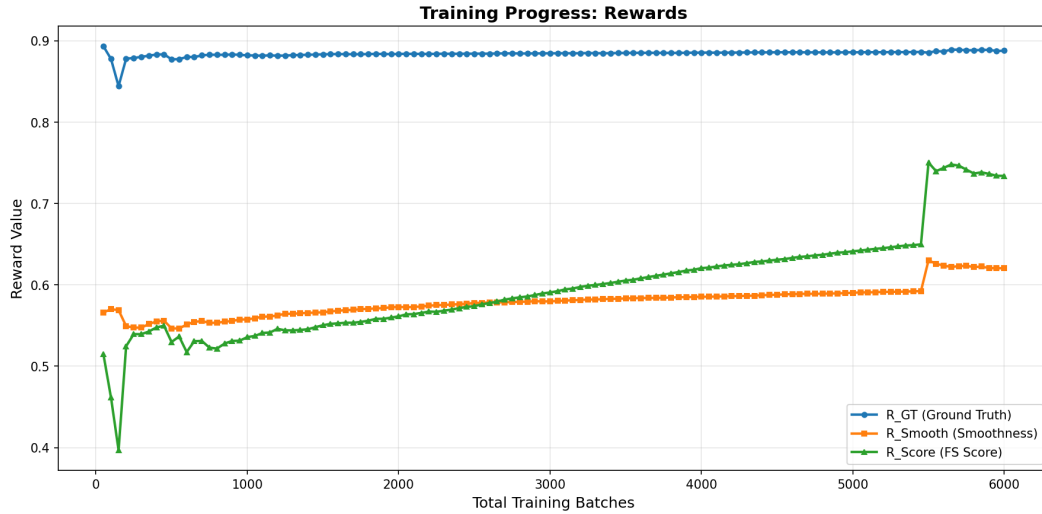


Figure 4: **Reward over Batches.** Training curves for the vis-GRPO stage over 6,000 batches.

Figure 4 illustrates how the reward function evolves over 6,000 batches. The training curves demonstrate that the reinforcement learning process successfully balances multiple objectives: our model maintains the ground-truth reward while the scoring reward grows gradually across batches. This behavior suggests the policy is effectively exploring the semantic space to satisfy language instructions while adhering to the physical constraints and motion priors established in the supervised stage.

670 I Qualitative Analysis

671 I.1 Metric Inconsistency: MPJPE vs. Biomechanical Quality

672 While Mean Per Joint Position Error (MPJPE) is the standard benchmark in motion generation, our
673 qualitative analysis reveals a significant divergence between lower MPJPE scores and actual biome-
674 mechanical validity in elite athletic scenarios. As demonstrated in Fig. 5 and Fig. 6, traditional metrics
675 often penalize high-quality, physically plausible motions while rewarding "static" or "averaged"
676 failure modes of baseline models.

677 I.1.1 The Phase-Shift Paradox and Static Risk-Aversion

678 In high-velocity rotational movements, such as the triple jumps (3A, 3S, 3LZ) shown in our qualitative
679 results, MPJPE is extremely sensitive to temporal alignment. We identify two primary failures of this
680 metric:

- 681 • **Spatio-temporal Misalignment:** Even if a model generates a physically perfect rotation
682 that follows the laws of angular momentum, a minor phase shift (timing difference) in the
683 rotational velocity relative to the ground truth (GT) results in massive Euclidean distances
684 between joints. Consequently, a correct but slightly faster/slower rotation is penalized more
685 heavily than an anatomically collapsed but spatially "near" posture.
- 686 • **Baseline "Static" Strategy:** Baselines like CoMusion and Transfusion exhibit a "risk-
687 aversion" behavior to minimize MPJPE. In the most complex aerial phases (Fig. 5), these
688 models often "freeze" the posture or skip directly to the landing preparation. While this
689 minimizes coordinate displacement by staying near the dataset average, it results in a total
690 loss of kinetic energy and biomechanical logic. Expert evaluations from professional coaches
691 and skaters confirmed that such "static" motions are physically impossible and lack the
692 dynamic chain required for successful jumps.

693 I.1.2 Fidelity vs. Strategic Optimization

694 Our results suggest that ReasonMotion treats motion generation not merely as an imitation task,
695 but as a strategic optimization problem. This is evidenced by the contrast between successful and
696 suboptimal reference motions:

- 697 • **High-Fidelity Reconstruction:** In cases where the original performance is elite (+GOE,
698 e.g., 3A_0402 in Fig. 5), ReasonMotion achieves high alignment with the GT. This proves
699 the model's capacity for precise reconstruction when the reference demonstrates a stable
700 kinetic chain.
- 701 • **Strategic Deviation for Correction:** In cases where the reference performance is flawed
702 (-GOE, e.g., Fig. 6), ReasonMotion intentionally deviates from the GT to seek a more
703 stable landing solution. By adjusting the axis of rotation or tightening the posture, the
704 model increases its MPJPE relative to the "stumbling" GT but produces a technically
705 superior "Idealized Form." This deviation represents a transition from mere imitation to
706 biomechanical consultation.

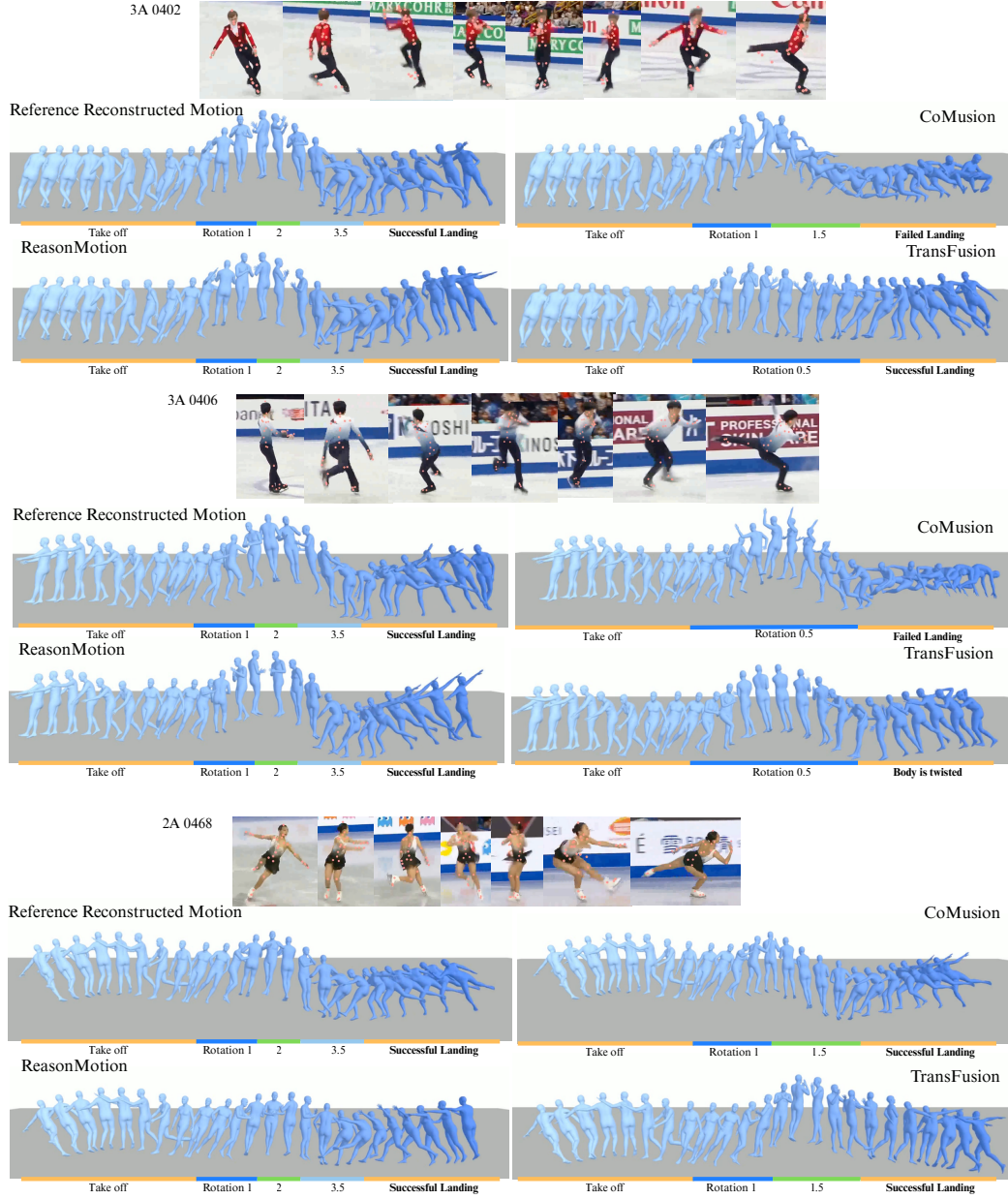


Figure 5: Qualitative analysis of successful jumps (+GOE). ReasonMotion maintains high fidelity to the elite reference while preserving fluid rotational momentum, whereas baselines often collapse or remain static during the peak flight phase.

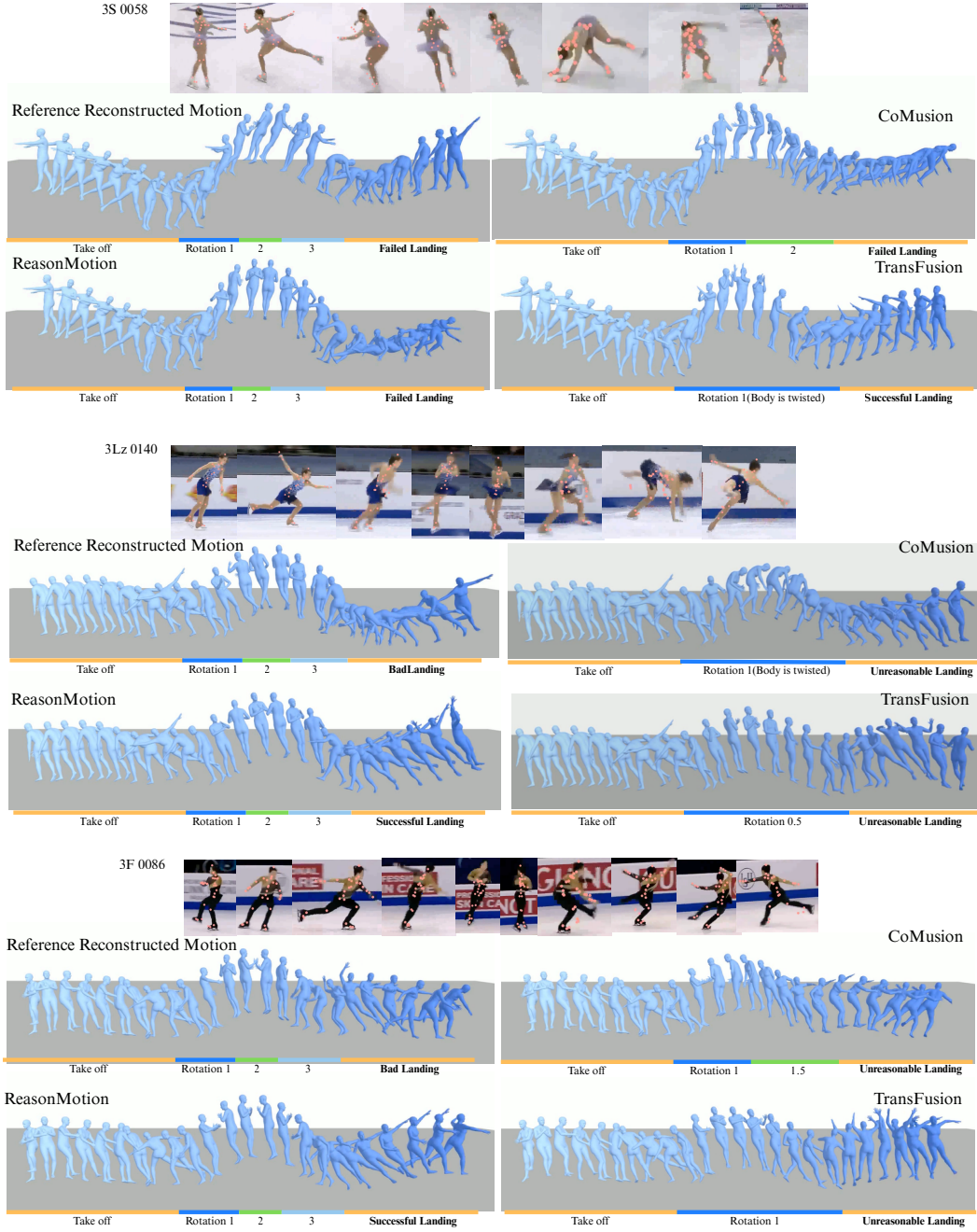


Figure 6: Qualitative analysis of suboptimal jumps (-GOE). In cases of failed references (e.g., 3S_0058), ReasonMotion attempts to optimize the biomechanical trajectory. However, it respects physical constraints, correctly identifying cases where a successful landing is physically unattainable due to poor takeoff parameters.

707 I.2 Emergent Learning Characteristics

708 **Evolution of Optimality (Fig. 7B).** We analyze how the model solves predicted failures through
 709 *vis*-GRPO, shifting from probabilistic realism to biomechanical optimality. *Realism (Early Stages):*
 710 Initially, the model reflects the athlete’s likely “Personal Style,” accurately predicting failure from a
 711 flawed takeoff. This serves as a risk assessment, showing the probable outcome if no corrections are
 712 made. *Optimality (Late Stages):* Through RL refinement, the model discovers extreme compensatory
 713 strategies (“Textbook Form”). It learns to save failing jumps via elite mid-air positional adjustments,
 714 providing a theoretical upper bound for successful landings despite suboptimal initial conditions.

715 **Semantic Controllability (Fig. 7C).** Using a fixed takeoff (Batch 2500), we vary the prompt from
 716 “Single” to “Quadruple.” The model predicts successful outcomes for 1–3 rotations but faithfully
 717 identifies that the takeoff velocity cannot support a Quadruple, resulting in a fall. This demonstrates
 718 the model does not function as a “physics-defying” animator. Despite attempting to correct the
 719 rotational axis and optimize the vertical kinetic chain, the model still simulates a fall if the takeoff
 720 conditions (e.g., initial angular momentum or center-of-mass trajectory) are beyond the physical
 721 threshold for recovery. This highlights the model’s utility as a biomechanical “what-if” simulator,
 722 accurately reflecting the limits of athletic recovery under suboptimal conditions.

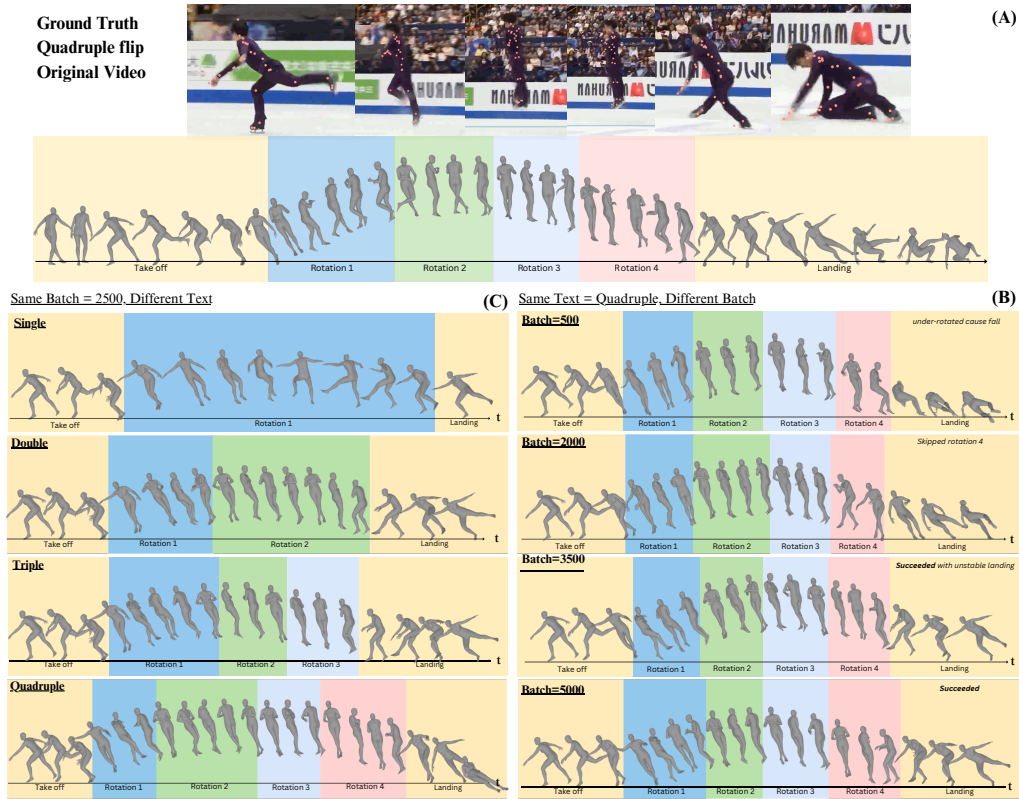


Figure 7: *Learning Evolution*. (A) Inference ground truth (a failed jump). (B) Optimal Recovery: By Batch 5000, the model discovers compensatory adjustments to “save” the landing. (C) Physical Honesty: At Batch 2500, the model correctly predicts failure (fall) for an insufficient Quadruple takeoff.

723 I.3 Visual Fidelity and Biomechanical Realism

724 To evaluate technical adjustments, we utilize an instruction-grounded module for localized edits as
 725 interventions on the takeoff (Fig. 8). This “coaching sandbox” reveals how perturbations propagate
 726 through the kinetic chain. Unlike baselines (Transfusion) that yield “hunched” postures due to
 727 supervised dataset averaging, *vis*-GRPO prioritizes performance rewards to discover a stable, vertical
 728 kinetic chain. This biomechanical realism confirms that reward-guided reasoning produces elite-level
 729 motions, providing a reliable tool for risk assessment and strategy discovery.

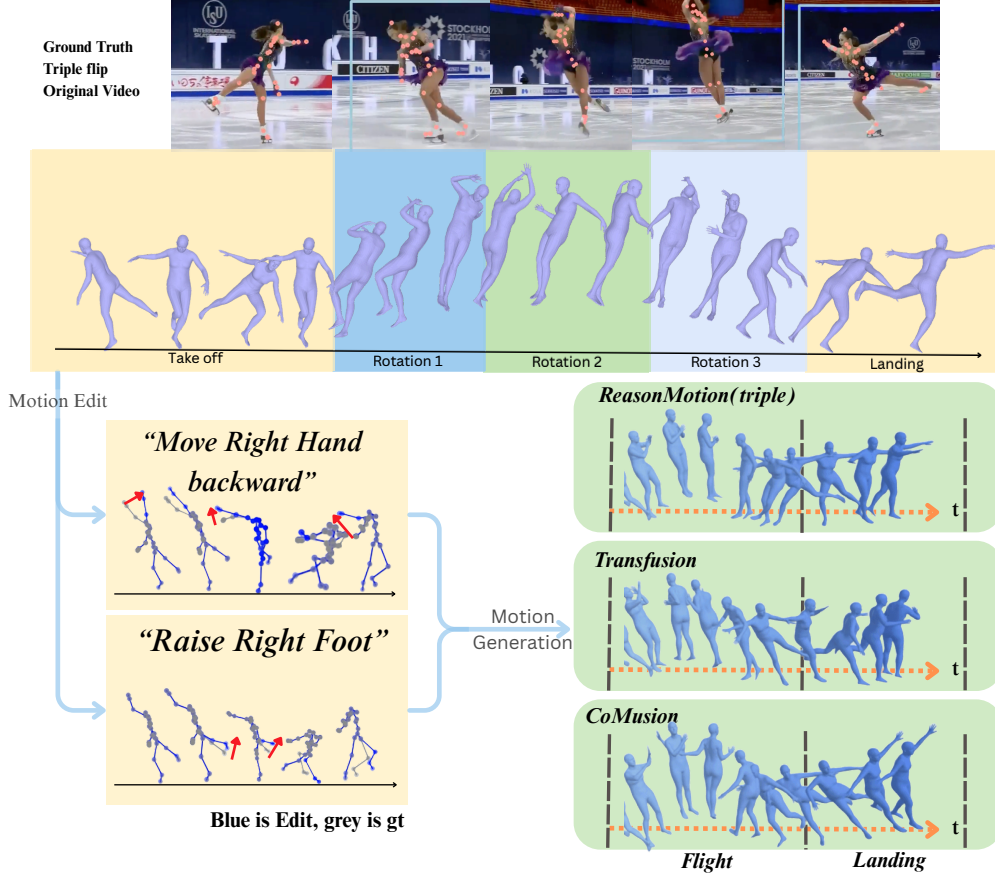


Figure 8: *Visual Comparison*. Top: Inference ground truth. Bottom Left: Localized edits on a takeoff (blue indicates edits, grey is ground truth). Bottom Right: Comparison. ReasonMotion maintains a vertical axis, whereas baselines produce unstable artifacts.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The main claims in the abstract and introduction accurately reflect the contributions and scope of this paper. We propose ReasonMotion, a framework that leverages Spatial-Temporal-Aware Mixture-of-Experts (ST-MoE) and visual Group Relative Policy Optimization (vis-GRPO) for instruction-based motion generation and refinement. To the best of our knowledge, ReasonMotion is the first framework designed to generate athlete motion as instructional demonstrations, enabling fine-grained, instruction-based refinement of sports techniques. All claims are supported by quantitative experiments on Human3.6M, HumanFineFS, and MotionFix, as well as qualitative validation by professional athletes.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: We discuss our limitation in the section 6, including the misalignment between automatic evaluation metrics and perceptual motion quality.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best

judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Yes, the formula for ST-MoE is articulated in section 3.2.1, and the formula for vis-GRPO is articulated in 3.2.2. Moreover, formula for the R3 algorithm used for combining ST-MoE and vis-GRPO is sketched in Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Yes, we provide all information necessary to reproduce our results. The full algorithm for training ST-MoE with vis-GRPO is detailed in Appendix A. The complete pipeline for training ReasonMotion is described in Section 3.3 and Section 3.4. All evaluations are conducted on publicly available datasets (MotionFix, Human3.6M, and HumanFineFS). Experimental setup and implementation details, including hyperparameters, are provided in Section 4.2 and Appendix G.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.

- (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
- (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The code is provided in the supplementary material, including scripts to reproduce the main experimental result.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: We provide these in Appendix G.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report standard deviation for the reward model used in vis-GRPO in table 3, as it is used to compute the GOE score for each generated motion. For other evaluation metrics (e.g., MPJPE, R@k, ADE) computed on fixed test splits of Human3.6M, Human-FineFS, and MotionFix, we follow standard practice in the motion generation literature and report single-run results, as variability across runs is minimal given the deterministic evaluation protocol.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: We report all compute details in Appendix G, including GPU model (NVIDIA RTX 4090) and per-experiment training times.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: This research fully conforms with the NeurIPS Code of Ethics. Our work is methodological in nature, proposing a new framework for motion reasoning trained and evaluated on publicly available datasets. No sensitive personal data was collected. Domain experts (professional athletes) were consulted to validate the qualitative correctness of our results, this involvement was limited to evaluation feedback and raises no ethical concerns. We foresee no significant potential for misuse or harm arising from this work.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The paper discusses potential societal impacts of this work. On the positive side, our method has direct applicability to sports coaching and athletic performance analysis, enabling more objective and scalable assessment of complex human motion. On the negative side, we do not identify any direct or foreseeable negative societal impacts, as the work is methodological in nature, operates on public benchmarks, and is not designed for generative or surveillance applications.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This paper poses no high risk for misuse. We do not release a pre-trained generative model, scrape or publish a new dataset, or produce artifacts that could be directly weaponized for harmful purposes. The method operates on existing public benchmarks, and no additional safeguards are required.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring

994 that users adhere to usage guidelines or restrictions to access the model or implementing
995 safety filters.
996 • Datasets that have been scraped from the Internet could pose safety risks. The authors
997 should describe how they avoided releasing unsafe images.
998 • We recognize that providing effective safeguards is challenging, and many papers do
999 not require this, but we encourage authors to take this into account and make a best
1000 faith effort.

1001 12. Licenses for existing assets

1002 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1003 the paper, properly credited and are the license and terms of use explicitly mentioned and
1004 properly respected?

1005 Answer: [\[Yes\]](#)

1006 Justification: We use three publicly available datasets and properly credit their original
1007 creators via citations. MotionFix [1] and HumanFineFS [35] are used under their respective
1008 open-source licenses. Human3.6M [12] is a widely used benchmark dataset available for
1009 non-commercial research purposes. All datasets are used strictly in accordance with their
1010 terms of use, and no redistribution or re-packaging is performed.

1011 Guidelines:

- 1012 • The answer [\[N/A\]](#) means that the paper does not use existing assets.
- 1013 • The authors should cite the original paper that produced the code package or dataset.
- 1014 • The authors should state which version of the asset is used and, if possible, include a
1015 URL.
- 1016 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1017 • For scraped data from a particular source (e.g., website), the copyright and terms of
1018 service of that source should be provided.
- 1019 • If assets are released, the license, copyright information, and terms of use in the
1020 package should be provided. For popular datasets, [paperswithcode.com/datasets](#)
1021 has curated licenses for some datasets. Their licensing guide can help determine the
1022 license of a dataset.
- 1023 • For existing datasets that are re-packaged, both the original license and the license of
1024 the derived asset (if it has changed) should be provided.
- 1025 • If this information is not available online, the authors are encouraged to reach out to
1026 the asset’s creators.

1027 13. New assets

1028 Question: Are new assets introduced in the paper well documented and is the documentation
1029 provided alongside the assets?

1030 Answer: [\[Yes\]](#)

1031 Justification: We release the implementation code as part of the supplementary material.
1032 The code includes all components necessary to reproduce the main results of the paper. No
1033 new datasets or pretrained model weights are released as part of this submission.

1034 Guidelines:

- 1035 • The answer [\[N/A\]](#) means that the paper does not release new assets.
- 1036 • Researchers should communicate the details of the dataset/code/model as part of their
1037 submissions via structured templates. This includes details about training, license,
1038 limitations, etc.
- 1039 • The paper should discuss whether and how consent was obtained from people whose
1040 asset is used.
- 1041 • At submission time, remember to anonymize your assets (if applicable). You can either
1042 create an anonymized URL or include an anonymized zip file.

1043 14. Crowdsourcing and research with human subjects

1044 Question: For crowdsourcing experiments and research with human subjects, does the paper
1045 include the full text of instructions given to participants and screenshots, if applicable, as
1046 well as details about compensation (if any)?

Answer: [Yes]

Justification: This paper involves research with human subjects. We conducted a user study in which professional athletes evaluated the quality and plausibility of our generated motions via a structured questionnaire. The full text of the questionnaire is provided in Appendix D. Participation was voluntary, and athletes were not compensated financially as they participated as domain expert collaborators. No crowdsourcing platform was used.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [Yes]

Justification: Our user study protocol was reviewed and approved by the Institutional Review Board (IRB) of our home institution (Protocol Number: 11501HT012) via an expedited review process. The study involved participants evaluating the perceptual quality of generated motions through a structured questionnaire, which posed minimal risk. All participants were fully informed of the study's purpose and the nature of the tasks, and they provided their consent prior to participation.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: Our paper does not use LLMs as an important, original, or non-standard component of the core methodology. Our method applies vis-GRPO to a motion diffusion model, and no language model is involved in the core pipeline.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.